In-Class Assignment: Logistic Regression

Please answer the following questions.
Code : please refer to the Copy of LogisticRegression.ipynb
Dataset: please refer to the https://www.kaggle.com/datasets/uciml/breast-cancer-wisconsin-data

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Breast Cancer Detection using Logistic Regression

Objective:
In this assignment, you will build a diagnostic model to classify tumors as either Malignant (1) or Benign (0) using the Breast Cancer Wisconsin Diagnostic dataset from Kaggle.

Dataset Overview:
Each row represents a patient's biopsy results. Features are computed from a digitized image of a fine needle aspirate (FNA) of a breast mass.
They describe characteristics of the cell nuclei present in the image.

Steps to Follow:

1. Data Loading: Use files.upload() to upload the data.csv file you downloaded from Kaggle. Use pandas to read the CSV file.

2. Data Cleaning (Preprocessing): Remove the id column as it does not contribute to prediction. Remove any empty columns (like Unnamed: 32). Encode the diagnosis column: Convert labels 'M' (Malignant) and 'B' (Benign) into numerical values (1 and 0).

3. Data Splitting: Split your dataframe into features (X) and the target variable (y). Then, use train_test_split to create training and testing sets (typically an 80/20 split).

4. Model Training: Initialize a LogisticRegression model. Note: You may need to increase max_iter (Ex: to 10000) to ensure the model converges.

5. Prediction & Evaluation: Use the test set to generate predictions and evaluate your model using a Confusion Matrix, Accuracy, Precision, and Recall.